why is it still not done?

digitalisierung und programmierung · prof. dr. nicolas meseth

1 where the time goes
2 three levers
3 what it costs to be sure

part 1

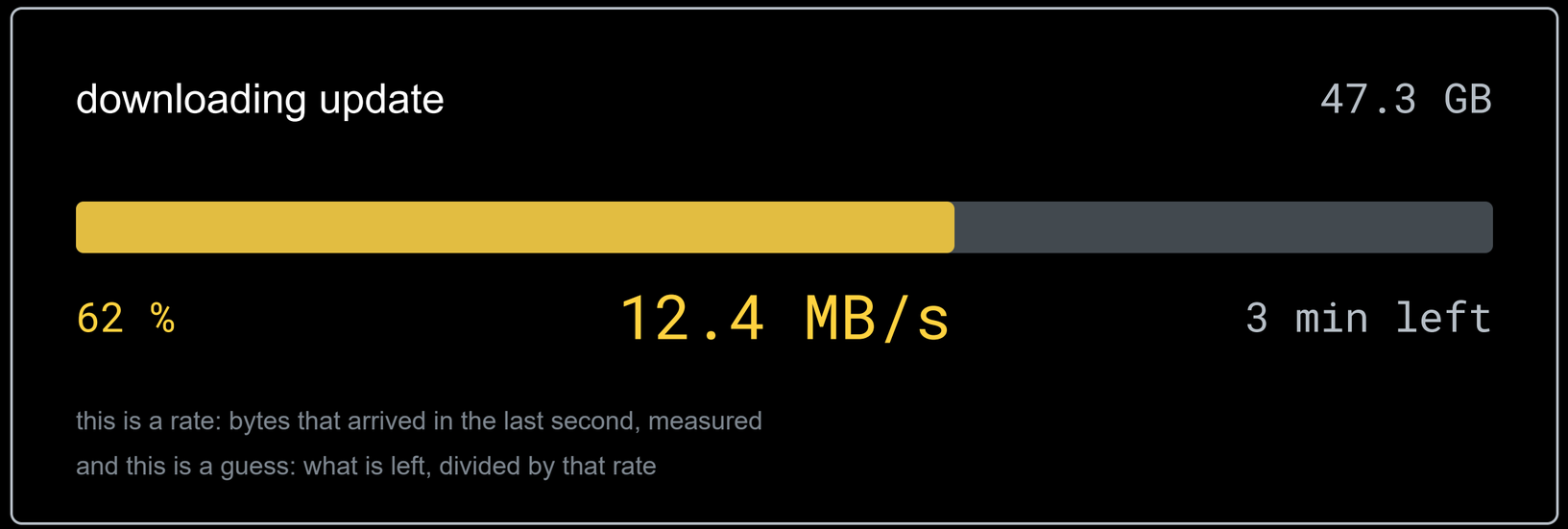
where the time goes

two kilobytes.
twenty-seven minutes.



image: ai-generated (gpt-image-2)

you know this screen



the rate is measured. the minutes are a guess made from it, and they jump when it does.

you know this screen



the rate is measured. the minutes are a guess made from it, and they jump when it does.

gross and net



nothing was lost. most of it was administration.

where the 24 went



deductions multiply. they do not add up.

where the 24 went



deductions multiply. they do not add up.

where the 24 went



deductions multiply. they do not add up.

where the 24 went



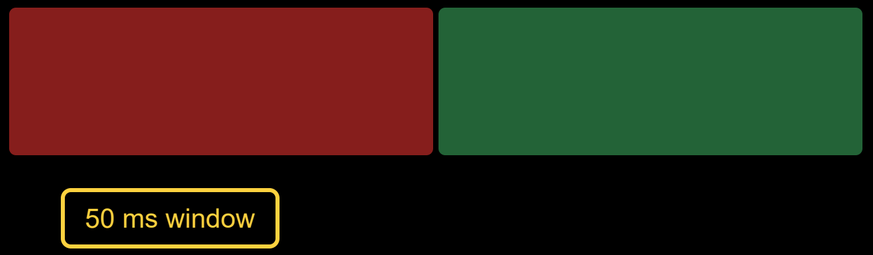
25 % + 40 % = 65 %, so 8.4?

no. the second cut only applies to what the first left over.

deductions multiply. they do not add up.

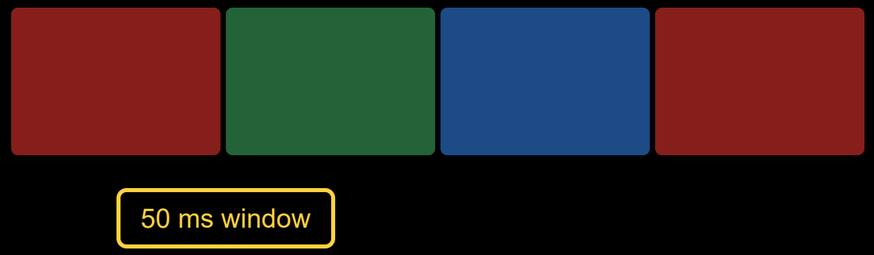
pulling the wrong lever

10 symbols per second: a symbol is 100 ms



20 bit/s, and all of it readable

20 symbols per second: a symbol is 50 ms



40 bit/s on paper, nothing readable

a lever past the bottleneck does not help. it destroys.

your limit is a measurement

not the sensor's datasheet: it does not know your room
not the best team's number: their setup is not yours
not the arithmetic with the values you wish for

what your link really does

team	net bit/s	how it was measured
team 3	21	50 messages, raised the rate until the first error
team 9	14	same, but by the window in the afternoon
team 6	12	estimated, not measured
team 1	31	with a cardboard tube around the sensor

same hardware. different rooms. different numbers.

part 2

three levers

```
t = n / (r * b)
```

a faster computer is not one of them.

$$t = n / (r * b)$$

b more bits per symbol

costs safety margin: narrower bands

a faster computer is not one of them.

$$t = n / (r * b)$$

b — more bits per symbol

costs safety margin: narrower bands

r — more symbols per second

costs the measurement window

a faster computer is not one of them.

$$t = n / (r * b)$$

b — more bits per symbol

costs safety margin: narrower bands

r — more symbols per second

costs the measurement window

n — fewer bits in total

costs a few milliseconds of computing

a faster computer is not one of them.

the same 20 bit/s, twice

team a

10 * 2

10 symbols per second, 4 colours

20 bit/s

close to the measurement window

team b

5 * 4

5 symbols per second, 16 colours

20 bit/s

close to the scatter limit

the product decides. the two factors decide what breaks first.

the third lever has its own deck

fewer bits in total: compression
costs milliseconds, not safety margin
works before the link, so it works on any link

next: compression

part 3

what it costs to be sure

three levers, one file

as it is  27 min

16 384 bits, 5 symbols per second

the levers are independent. that is why they multiply.

three levers, one file



16 384 bits, 5 symbols per second

the levers are independent. that is why they multiply.

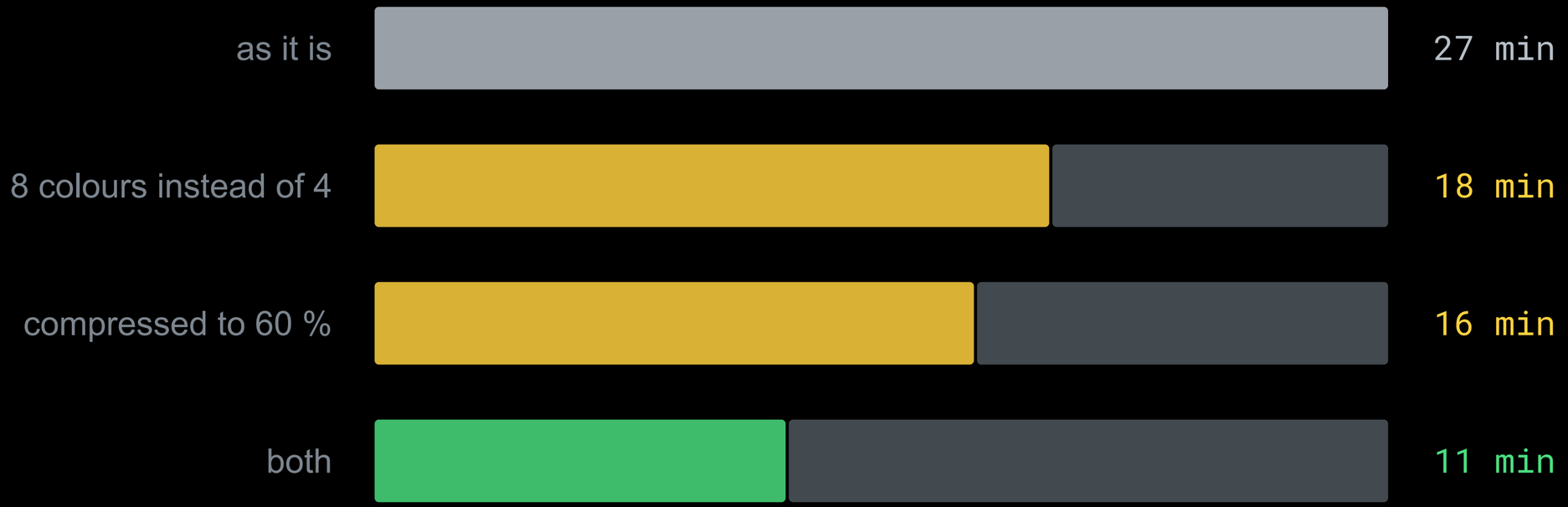
three levers, one file



16 384 bits, 5 symbols per second

the levers are independent. that is why they multiply.

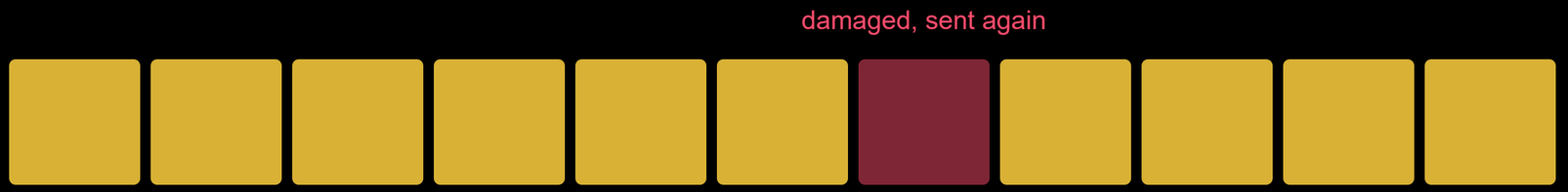
three levers, one file



16 384 bits, 5 symbols per second

the levers are independent. that is why they multiply.

every tenth one again



11 transmissions

10 messages delivered

10 / 11 = 0.91

not 0.90

nothing was lost. the link was busy anyway.

acknowledge each, or a block

acknowledge every message



acknowledge every block of ten



yellow: payload · dark: waiting for the acknowledgement

but one damaged message costs the whole block again

how big a block? that depends on a number you have to measure.

throughput is not delay

20 bit/s does not mean a one-byte command is there after 0.4 seconds. around it sit the frame, the recognising and the reaction.

1 measure your own net rate, and write down how
2 find the lever that is cheapest on your setup, with the number that says so
3 pick a block size for your acknowledgements, with a reason

the neck decides
how much gets through.
you decide
what you send.



image: ai-generated (gpt-image-2)